Calculus III Cheat Sheet

§1 3D Space

Equations of Lines

$$\begin{aligned}
\text{Vector form}: &\quad \vec{r} = \vec{r_0} + t\vec{v} = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle \\
\text{Parametric form}: &\quad x = x_0 + ta, y = y_0 + tb, z = z_0 + tc \\
\text{Symmetric form}: &\quad \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}
\end{aligned}$$

Equations of Planes

Given a normal vector $\vec{n} = \langle a, b, c \rangle$ and a point on the plane $\vec{r_0}$, we have

$$\begin{aligned}
\text{Vector equation:} &\quad \vec{n} \cdot (\vec{r} - \vec{r_0}) = 0 \\
\text{Scalar equation:} &\quad ax + by + cz = d
\end{aligned}$$

Quadric Surfaces

A *quadric surface* is the graph of an equation in this form:

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

Some common examples:

$$\begin{aligned}
\text{Ellipsoid: } & \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \\
\text{Cone: } & \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2} \\
\text{Cylinder: } & \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\
\text{Hyperboloid of one sheet: } & \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \\
\text{Hyperboloid of two sheets: } & -\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \\
\text{Elliptic Paraboloid: } & \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} \\
\text{Hyperbolic Paraboloid: } & \frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}
\end{aligned}$$

Functions of Several Variables

One way to visualize multivariable functions is to take *level sets* (sometimes called *level curves*, *level surfaces*, *contour curves*, etc. when appropriate), where you set one variable constant and see what happens with the others. These can also be called *traces* when x or y is what you're holding constant.

Vector-Valued Functions

A *vector-valued function* (or *vector function*) is a function which takes in one or more variables and returns a vector. A single-variable vector-valued function is often expressed like $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, where f, g, and h are called *component functions*.

Calculus with Vector-Valued Functions

A *smooth curve* is a curve $\vec{r}(t)$ for which $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$ for all t except possibly at the endpoints.

Some useful facts (framed in $\mathbb{R}^3$, but can be generalized to $\mathbb{R}^n$):

$$\begin{aligned}
\lim_{t\to a} \langle f(t), g(t), h(t) \rangle &= \left\langle \lim_{t\to a} f(t), \lim_{t\to a} g(t), \lim_{t\to a} h(t) \right\rangle \\
\frac{d}{dt} \langle f(t), g(t), h(t) \rangle &= \langle f'(t), g'(t), h'(t) \rangle \\
\int \langle f(t), g(t), h(t) \rangle \, dt &= \left\langle \int f(t)dt, \int g(t)dt, \int h(t)dt \right\rangle \\
\int_a^b \langle f(t), g(t), h(t) \rangle \, dt &= \left\langle \int_a^b f(t)dt, \int_a^b g(t)dt, \int_a^b h(t)dt \right\rangle .
\end{aligned}$$

Tangent, Normal, and Binormal Vectors

Given a function $\vec{r}: \mathbb{R} \to \mathbb{R}^3$, the *unit tangent vector* is defined by

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||}.$$

The *unit normal vector* is orthogonal to the unit tangent vector and is defined by

$$\vec{N}(t) = \frac{\vec{T}'(t)}{\left|\left|\vec{T}'(t)\right|\right|}.$$

The *binormal vector* is orthogonal to both the unit tangent vector and the unit normal vector and is defined by

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t).$$

Arc Length with Vector Functions

Arc length of a single-variable, vector-valued function $\vec{r}$:

$$L = \int_a^b ||r'(t)||\, dt.$$

The *arc length function* is defined as

$$s(t) = \int_0^t ||r'(u)||\, du,$$

which can be useful when we want to reparametrize the function in terms of arc length by solving the resulting $s(t)$ for t.

Curvature

The *curvature* of a function is defined as

$$\kappa = \left|\left|\frac{d\vec{T}}{ds}\right|\right|.$$

Here are two alternate forms that are often somewhat easier to compute:

$$\kappa = \frac{\left|\left|\vec{T}'(t)\right|\right|}{||r'(t)||} \qquad \kappa = \frac{||r'(t) \times r''(t)||}{||r'(t)||^3}$$

Velocity and Acceleration

It is common to break up acceleration into its tangential and normal components (a_T and a_N, respectively). Letting $\vec{v}$ be the velocity and $v = ||\vec{v}(t)||$, these have simple formulas:

$$a_T = v' = \frac{r'(t) \cdot r''(t)}{||r'(t)||} \qquad a_N = \kappa v^2 = \frac{||r'(t) \times r''(t)||}{||r'(t)||}$$

Cylindrical Coordinates

Cylindrical coordinates are in the form (r, θ, z), where r and θ act just as in polar coordinates, and the z acts like the usual Cartesian z (refer to Figure 1 below).

Conversion formulas

Cylindrical to Cartesian:

$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ z &= z \end{aligned}$$

Cartesian to cylindrical:

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ \theta &= \tan^{-1}\left(\frac{y}{x}\right) \\ z &= z \end{aligned}$$

Spherical Coordinates

Spherical coordinates are of the form (ρ, θ, φ), where:

- ρ is the distance from the origin to the point (so $\rho \geq 0$).
- θ is the same as θ from polar/cylindrical coordinates.
- φ is the angle between the positive z-axis and the line from the origin to the point. We restrict to $0 \leq \varphi \leq \pi$.

Conversion formulas

Spherical to cylindrical:

$$\begin{aligned} r &= \rho\sin\varphi \\ \theta &= \theta \\ z &= \rho\cos\varphi \end{aligned}$$

Spherical to Cartesian:

$$\begin{aligned} x &= \rho\sin\varphi\cos\theta \\ y &= \rho\sin\varphi\sin\theta \\ z &= \rho\cos\varphi \end{aligned}$$

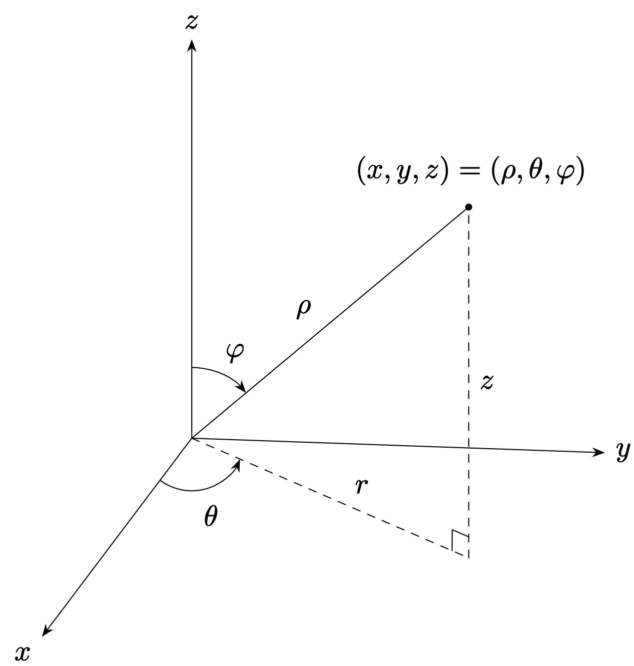

Figure 1: A comparison of spherical and cylindrical coordinates

§2 Partial Derivatives

Partial Derivatives

To take a derivative with respect to one variable, just take the normal derivative with respect to that variable, holding all other variables constant.

For higher-order partial derivatives, note that we have $f_{xy} = f_{yx}$ at a point whenever both mixed partials are continuous on some neighborhood of that point. This is Clairaut's theorem.

Differentials

The *differential* or *total differential* of a two-variable function $z = f(x, y)$ is defined as

$$dz = df = f_x dx + f_y dy.$$

This extends naturally to functions of more variables.

Chain Rule

The chain rule for a function z of n variables $x_1, x_2, \ldots, x_n$ which are each functions of m variables $t_1, t_2, \ldots, t_m$ is defined as

$$\frac{\partial z}{\partial t_i} = \frac{\partial z}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial z}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial z}{\partial x_n}\frac{\partial x_n}{\partial t_i}.$$

Directional Derivatives

We define the *gradient* of a three-variable function f to be

$$\nabla f = \langle f_x, f_y, f_z \rangle .$$

Now given any unit vector $\vec{u}$, we say that the *directional derivative* of f in the direction of $\vec{u}$ is

$$D_{\vec{u}} f = \nabla f \cdot \vec{u}.$$

A couple of theorems about gradients:

1. The maximum value of $D_{\vec{u}} f(\vec{x})$ is given by $||\nabla f(\vec{x})||$ and will occur in the direction of $\nabla f(\vec{x})$.
2. The gradient vector $\nabla f(x_0, y_0)$ is orthogonal to the level curve $f(x, y) = k$ at the point (x_0, y_0).

§3 Applications of Partial Derivatives

Tangent Planes

The equation of the tangent plane to a graph $f(x, y, z) = k$ at the point (x_0, y_0, z_0) is given by

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0,$$

which you can remember because it's ensuring that points on the plane, which are of the form $(x - x_0, y - y_0, z - z_0)$, are orthogonal to the gradient, so the equation becomes

$$(\nabla f)(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0.$$

Relative Minimums and Maximums

The point (a, b) is a *critical point* of $f(x, y)$ provided one of the following is true:

1. $\nabla f(a, b) = \vec{0}$
2. $f_x(a, b)$ and/or $f_y(a, b)$ doesn't exist.

If (a, b) is a relative extrema of $f(x, y)$ and the first order derivatives of f exist at (a, b), then (a, b) is a critical point and $\nabla f(a, b) = \vec{0}$.

Let (a, b) be a critical point of $f(x, y)$ where the second order partials are continuous around (a, b). Define

$$D = D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2 .$$

Then

1. If $D > 0$, and $f_{xx}(a, b) > 0$ then there is a relative minimum at (a, b)
2. If $D > 0$, and $f_{xx}(a, b) < 0$ then there is a relative maximum at (a, b)
3. If $D < 0$, then the point (a, b) is a saddle point
4. If $D = 0$, then (a, b) may be any of the three types.

Absolute Minimums and Maximums

Extreme Value Theorem: if $f(x, y)$ is continuous on a closed and bounded set D, then it achieves an absolute maximum and an absolute minimum on D.

Method for finding absolute extrema on a closed region D:

1. Find all the critical points.
2. Find extrema along the boundary (reduces to a Calc I problem).
3. The largest is the absolute maximum and the smallest is the absolute maximum.

Lagrange Multipliers

This is another way to optimize, called the Method of Lagrange Multipliers:

1. Solve the following system of equations:

$$\begin{aligned}\nabla f(x, y, z) &= \lambda \nabla g(x, y, z)\\ g(x, y, z) &= k.\end{aligned}$$

2. Plug all solutions from the first step into f and identify the minimum and maximum values, provided they exist and $\nabla g \neq \vec{0}$ at the point. (The constant λ is the *Lagrange Multiplier*.)
3. Look for any remaining solutions along the boundary.

§4 Multiple Integrals

Double Integrals

The definition of a double integral over a rectangular region R:

$$\iint_R f(x, y)\, dA = \lim_{n,m\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_i^*, y_j^*)\Delta A,$$

where x_i^* and y_j^* are points from somewhere within the subdivisions (can be just midpoints).

Iterated Integrals

Fubini's Theorem: If $f(x, y)$ is continuous on $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy.$$

Helpful trick: if $f(x, y) = g(x)h(y)$ and we are integrating over $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \left(\int_a^b g(x)\, dx\right) \left(\int_c^d h(y) dy\right).$$

Note that if we take the indefinite integral of a multivariable function with respect to one variable, the "constant" of integration is a function of the other variables.

Double Integrals over General Regions

Double Integrals in Polar Coordinates

Triple Integrals

Triple Integrals in Cylindrical Coordinates

Triple Integrals in Spherical Coordinates

Change of Variables

Surface Area

Area and Volume Revisited

§5 Line Integrals

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Vector Fields

Line Integrals — Part I

Line Integrals — Part II

Line Integrals of Vector Fields

Fundamental Theorem for Line Integrals

Conservative Vector Fields

Green's Theorem

§6 Surface Integrals

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Curl and Divergence

Parametric Surfaces

Surface Integrals

Surface Integrals of Vector Fields

Stokes' Theorem

Divergence Theorem